Lefkios Mavroudi

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LinkedIn | GitHub | Website

Education

Imperial College London — *London, UK*
MSc Computing (Artificial Intelligence and Machine Learning) — *Sep 2026 – Sep 2027 (expected)*
- Safe Bulkers Scholarship (1 of 2 postgraduate awards, 2026–27).

University of Cyprus — *Nicosia, Cyprus*
BEng Electrical and Computer Engineering — *2022 – 2026*
- GPA 8.46/10; ranked 2nd of 39 in Computer Engineering and 4th of 83 in ECE; departmental award for 2nd-highest academic performance in year (2023–24, 2024–25, 2025–26).
- Thesis (10/10; Best Senior Design Project award): adversarial robustness of edge-deployed object detectors for mobile robots.

Experience

CYENS Centre of Excellence — *Jun 2026 – Jul 2026*
AI Research Engineer Intern — *Nicosia, Cyprus*
- Proposed and built a PyTorch pipeline on a SLURM GPU cluster that turns a transferred segmentation model's predictions into bounding-box prompts for MedSAM2, for right-ventricle segmentation on 204 EchoNet-Dynamic echocardiography cases.
- Ran 12 decoder fine-tuning variants (staged and adaptive unfreezing, prompt jitter, hard-example sampling), selected on validation only and scored a locked 41-case set once: mean Dice 0.8463 $\rightarrow$ 0.8918, with 74 of 82 frames improved.

KIOS Research and Innovation Centre of Excellence — *Jun 2025 – Jun 2026*
Researcher, Adversarial ML and Edge AI — *Nicosia, Cyprus*
- Built a PyTorch Expectation-over-Transformation patch attack across 9 CARLA-GeAR scenarios: YOLOv8n mAP@0.50 fell from 80.6% to 49.2% and high-confidence false positives rose $\sim$40$\times$, with black-box transfer to YOLOv5n, YOLOv8s and RT-DETR-L.
- Profiled the full pipeline on NVIDIA Jetson AGX Orin: the attack added only 2.4 ms end-to-end latency (+3.5%) and $-$0.05 W, so latency and power telemetry could not flag it. Of 5 preprocessing defences, Local Gradient Smoothing cut false-positive flooding by 63.6%.
- **First author and presenter**, *RAPTOR: Robustness Assessment of Patch Transfer and Operational Risk in Edge Object Detectors for Mobile Robots*, IEEE COINS 2026, Bologna.

Deloitte — *Jun 2024 – Jul 2024*
Cyber Risk Intern — *Nicosia, Cyprus*
- Performed ethical-hacking assessments and EU compliance exercises, documenting findings and prioritising remediation.

3AE Health — *Jun 2023 – Jul 2023*
AI Software Engineering Intern — *Nicosia, Cyprus*
- Co-built (team of 2) a traffic-analytics pipeline: YOLOv8s vehicle detection, multi-object tracking and 5-minute flow aggregation feeding a stacked-GRU forecaster (0.041 normalised validation MAE).

Projects

LLM Post-Training: Step-Level vs Outcome Rewards — *Aug 2026*
PyTorch | TRL (GRPO) | PEFT/LoRA | OpenAI API | W&B | GitHub
- Fine-tuned Qwen2.5-7B with GRPO and LoRA (0.13% of parameters trainable) on filtered GSM8K and MATH problems, comparing outcome rewards with step-level rewards from a PRM800K-trained classifier and from GPT-4o mini as an LLM judge.
- GSM8K (majority vote of 8, n=200): the PRM800K step reward reached 94.5% vs 91.0% baseline (paired McNemar $p = 0.04$); on MATH-500, all three GRPO reward types improved on the baseline by 4–5.5pp.

Adversarial Person-Hiding Patches — *Aug 2025 – Sep 2025*
PyTorch | YOLOv8 | GitHub
- Optimised printable patches that hide the holder from YOLOv8 (top-K suppression, TV/L2 regularisation) and ran them as a live webcam demo with switchable defences at Researchers' Night 2025; the demo video reached 183k LinkedIn impressions.

MIPS32 Processor Simulator — *Feb 2025 – Apr 2025*
C++ | Computer Architecture | GitHub
- Wrote cycle-accurate single-cycle, multi-cycle and pipelined MIPS32 simulators with hazard detection, forwarding and control-flow handling (graded 10/10).

Technical Skills

Languages: Python, C++, C, Java, SQL

ML and LLMs: PyTorch, Hugging Face Transformers, TRL/GRPO, PEFT/LoRA, OpenAI API, Ultralytics YOLO, OpenCV, scikit-learn, TensorFlow, MLX, PyTorch Geometric

Systems and Tools: Linux, Git, Docker, SLURM, Weights & Biases, NVIDIA Jetson AGX Orin, CARLA, Isaac Sim, ROS 2